

Differential Forms and Hodge Decomposition: An Analytic Approach

A Dissertation Submitted
in Partial Fulfilment of the Requirements
for the Degree of

MASTER OF SCIENCE

in

MATHEMATICS

by

Tharun N
(Roll No. IMS22252)



to

SCHOOL OF MATHEMATICS
INDIAN INSTITUTE OF SCIENCE EDUCATION AND
RESEARCH
THIRUVANANTHAPURAM - 695 551, INDIA

June 13, 2026

DECLARATION

I, **Tharun Naagesh (Roll No: IMS22252)**, hereby declare that, this report entitled “**Quantum Blackholes, Wall Crossing and Mock Modular Forms**” submitted to Indian Institute of Science Education and Research Thiruvananthapuram towards the partial requirement of **Master of Science in Mathematics**, is an original work carried out by me under the supervision of **Dr. Bindusar Sahoo** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold academic ethics and honesty. Whenever a piece of external information or statement or result is used then, that has been duly acknowledged and cited.

Thiruvananthapuram - 695 551
June 13, 2026

[Full Name]

CERTIFICATE

This is to certify that the work contained in this project report entitled “**My Project Title**” submitted by **[Full Name]** (Roll No: **[Roll Number]**) to Indian Institute of Science Education and Research, Thiruvananthapuram towards the partial requirement of **[Master of Science/ Doctor of Philosophy]** in **[Department Name]** has been carried out by [him/her/them] under my supervision and that it has not been submitted elsewhere for the award of any degree.

Thiruvananthapuram - 695 551
June 13, 2026

[Project Supervisor]
Project Supervisor

ACKNOWLEDGEMENT

Write about the people and the things you are indebted to in fulfilling this project

Lorem ipsum dolor sit amet, officia excepteur ex fugiat reprehenderit enim labore culpa sint ad nisi Lorem pariatur mollit ex esse exercitation amet. Nisi anim cupidatat excepteur officia. Reprehenderit nostrud nostrud ipsum Lorem est aliquip amet voluptate voluptate dolor minim nulla est proident.

I am thankful to Nikhil Alex Verghese, BS-MS'17, IISER Thiruvananthapuram for creating a 'Official IISER Thiruvananthapuram Thesis/Dissertation Format' in Overleaf.

Nostrud officia pariatur ut officia. Sit irure elit esse ea nulla sunt ex occaecat reprehenderit commodo officia dolor Lorem duis laboris cupidatat officia voluptate. Culpa proident adipisicing id nulla nisi laboris ex in Lorem sunt duis officia eiusmod. Aliqua reprehenderit commodo ex non excepteur duis sunt velit enim. Voluptate laboris sint cupidatat ullamco ut ea consectetur et est culpa et culpa duis.

Thiruvananthapuram - 695 551

[Full Name]

ABSTRACT

If you have to structure it as objective, methods, results, and conclusions, do it that way.

Lorem ipsum dolor sit amet, officia excepteur ex fugiat reprehenderit enim labore culpa sint ad nisi Lorem pariatur mollit ex esse exercitation amet. Nisi anim cupidatat excepteur officia. Reprehenderit nostrud nostrud ipsum Lorem est aliquip amet voluptate voluptate dolor minim nulla est proident. Nostrud officia pariatur ut officia. Sit irure elit esse ea nulla sunt ex occaecat reprehenderit commodo officia dolor Lorem duis laboris cupidatat officia voluptate. Culpa proident adipisicing id nulla nisi laboris ex in Lorem sunt duis officia eiusmod. Aliqua reprehenderit commodo ex non excepteur duis sunt velit enim. Voluptate laboris sint cupidatat ullamco ut ea consectetur et est culpa et culpa duis.

Contents

Contents	vi
List of Figures	vii
List of Tables	viii
1 Preliminaries	1
Appendices	3
A Some results from Calculus on $\mathbb{R}^n$	3
A.1 Differentiation Results	3
A.2 Integration Results	5
B Exterior Forms	7
B.1 Definitions and Results	7

List of Figures

List of Tables

Chapter 1

Preliminaries

Appendices

Appendix A

Some results from Calculus on $\mathbb{R}^n$

A.1 Differentiation Results

Lemma A.1.1. *There exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is positive on $(-1, 1)$, 0 elsewhere and is C^∞*

Proof. Consider

$$f := \begin{cases} e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} & \text{if } x \in (-1, 1) \\ 0 & \text{elsewhere} \end{cases}$$

This function is C^∞ and is positive on $(-1, 1)$ and 0 elsewhere.

□

Lemma A.1.2. *Let $a \in \mathbb{R}^n$, $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by*

$$g(x) = f([x^1 - a^1]/\varepsilon) f([x^2 - a^2]/\varepsilon) \cdots f([x^n - a^n]/\varepsilon)$$

g is then C^∞ , and has positive values only on $(a^1 - \varepsilon, a^1 + \varepsilon) \times (a^2 - \varepsilon, a^2 + \varepsilon) \cdots \times (a^n - \varepsilon, a^n + \varepsilon)$.

Theorem A.1.3. (*Inverse Function Theorem*) *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuously differentiable. Suppose that $\det Df(a) \neq 0$ for a given $a \in \mathbb{R}^n$. Then, there exists U and V open in $\mathbb{R}^n$ such that $a \in U$, $f(a) \in V$ and $f|_U : U \rightarrow V$ is a diffeomorphism.*

Theorem A.1.4. (Implicit Function Theorem) Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be continuously differentiable in an open set of (a, b) and let $f(a, b) = 0$. Let M be the $m \times m$ matrix

$$D_{n+j}f^i(a, b); 1 \leq i, j \leq m$$

and suppose $\det M \neq 0$. Then there exists open sets $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^m$ with $a \in A$, $b \in B$ such that for all $x \in A$ there is a unique $g(x) \in B$ so that $f(x, g(x)) = 0$, and the g so obtained is differentiable.

Proof. Define $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n \times \mathbb{R}^m$ given by $F(x, y) := (x, f(x, y))$. This function is clearly differentiable, and moreover,

$$DF = \begin{pmatrix} \frac{\partial x^1}{\partial x^1} & \frac{\partial x^1}{\partial x^2} & \cdots & \frac{\partial x^1}{\partial x^n} & \frac{\partial x^1}{\partial y^1} & \frac{\partial x^1}{\partial y^2} & \cdots & \frac{\partial x^1}{\partial y^m} \\ \frac{\partial x^2}{\partial x^1} & \frac{\partial x^2}{\partial x^2} & \cdots & \frac{\partial x^2}{\partial x^n} & \frac{\partial x^2}{\partial y^1} & \frac{\partial x^2}{\partial y^2} & \cdots & \frac{\partial x^2}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x^n}{\partial x^1} & \frac{\partial x^n}{\partial x^2} & \cdots & \frac{\partial x^n}{\partial x^n} & \frac{\partial x^n}{\partial y^1} & \frac{\partial x^n}{\partial y^2} & \cdots & \frac{\partial x^n}{\partial y^m} \\ \frac{\partial f^1}{\partial x^1} & \frac{\partial f^1}{\partial x^2} & \cdots & \frac{\partial f^1}{\partial x^n} & \frac{\partial f^1}{\partial y^1} & \frac{\partial f^1}{\partial y^2} & \cdots & \frac{\partial f^1}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1} & \frac{\partial f^m}{\partial x^2} & \cdots & \frac{\partial f^m}{\partial x^n} & \frac{\partial f^m}{\partial y^1} & \frac{\partial f^m}{\partial y^2} & \cdots & \frac{\partial f^m}{\partial y^m} \end{pmatrix}$$

And it is easy to see that $\det DF(a, b) = \det M \neq 0$. Therefore, from Inverse function theorem (with little modification), there exists an open set $A \times B$ of $\mathbb{R}^n \times \mathbb{R}^m$, and an open set $W \subset \mathbb{R}^n \times \mathbb{R}^m$ such that $(a, b) \in A \times B$, $(a, 0) \in W$ and $F : A \times B \rightarrow W$ is a diffeomorphism. $F^{-1} : W \rightarrow A \times B$ is also a diffeomorphism. Moreover, note that $F^{-1}(p, q)$ is of the form $(p, k(p, q))$ for some differentiable function $k : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$. $\pi \circ F(x, y) = \pi \circ (x, f(x, y)) = f(x, y)$ so $\pi \circ F \equiv f$. Look at $f \circ (p, k(p, q)) = f \circ F^{-1}(p, q) = \pi(p, q) = q$ so $f \circ (p, k(p, 0)) = 0$. Our $g(p)$ is given by $k(p, 0)$, and it is differentiable as a composition of differentiable functions.

□

Theorem A.1.5. (A corollary of A.1.4) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ ($p \leq n$) be continuously differentiable in an open set containing $a \in \mathbb{R}^n$. If $f(a) = 0$ and the matrix $D_j(f^i(a))_{p \times n}$ has rank p , then there exists an open neighbourhood A of a and a diffeomorphism $h : A \rightarrow \mathbb{R}^n$ such that $f \circ h(x_1, x_2, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$.

Proof. Consider $f : \mathbb{R}^{n-p} \times \mathbb{R}^p \rightarrow \mathbb{R}^p$. Suppose that the matrix $Q = D_{n-p+i}(f^j); i, j \leq p$ be invertible ($\det Q \neq 0$). Let, as before, $F : \mathbb{R}^{n-p} \times \mathbb{R}^p \rightarrow \mathbb{R}^{n-p} \times \mathbb{R}^p$ given by

$$F(p, q) = (p, f(p, q))$$

Since $\det DF \neq 0$, there exists, around $a = (a^1, a^2, \dots, a^{n-p}, a^{n-p+1}, \dots, a^n)$ a neighbourhood $A_{n-p} \times A_p \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ and a neighbourhood $W \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ containing $F(a) = (a_1, a_2 \dots a_{n-p}, 0)$ such that $F : A_{n-p} \times A_p \rightarrow W$ is invertible and whose inverse is continuously differentiable. Note that, the inverse of F ought to be of the form $F^{-1}(x_1, x_2, \dots, x_{n-p}, y_{n-p+1}, \dots, y_n) = (\vec{x}, b(\vec{x}, \vec{y}))$ (let $\vec{x} = (x_1, x_2, \dots, x_{n-p}) \in \mathbb{R}^{n-p}$ and $\vec{y} = (y_{n-p+1}, y_{n-p+2}, \dots, y_n) \in \mathbb{R}^p$) where it is understood that $b : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is continuously differentiable. Also note that $\pi_2 \circ F \equiv f$. We have that

$$f \circ F^{-1}(x_1, x_2 \dots x_{n-p}, x_{n-p+1}, \dots, x_n) =$$

$$\pi_2(x_1, x_2, \dots, x_{n-p}, x_{n-p+1}, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$$

Case 2: If the indices where the matrix $\{(\partial_i f^j(a)); 1 \leq j \leq p, i = i_1, i_2, \dots, i_p\}$ is not the last p indices, we write the linear transformation $\text{Swap} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that takes these p indices to the last p indices. This is a linear transformation that is invertible since it takes basis elements to basis elements. If we look then, at $f \circ \text{Swap} : \mathbb{R}^n \rightarrow \mathbb{R}^p$, we see that this is a function of the form of Case 1, so we have a h such that $f \circ \text{Swap} \circ h(x_1, x_2, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$ which means our desired function is $\text{Swap} \circ h$.

□

A.2 Integration Results

Definition A.2.1. (Partition) For closed interval $[a, b]$ in $\mathbb{R}$, a partition is simply a collection of sub-intervals $[a_i, a_{i+1}]$ so that $a_0 = a < a_1 < a_2 < \dots < a_k = b$.

For a closed rectangle R in $\mathbb{R}^n$, a partition P , usually written along with its subpartitions (as in, $P \equiv (P_1, P_2, \dots, P_n)$), is a collection of subrectangles R_i (of a particular form) so that $R = \cup_i R_i$ and $R_i^\circ \cap R_j^\circ = \emptyset$. Explicitly, if

$$R = [a^1, b^1] \times [a^2, b^2] \times \dots \times [a^n, b^n]$$

,

$$P_i \equiv [a^i = a_0^i < a_1^i < a_2^i \cdots a_{k_i}^i = b^i]$$

and I_j^i denotes the j -th subinterval of the i -th partition, the collection of all possible sub rectangles of R of the form $\{I_{j_1}^1 \times I_{j_2}^2 \times \cdots \times I_{j_n}^n : 1 \leq j_q \leq \text{number of intervals}(P_q)\}$ is a partition of R .

Definition A.2.2. (Upper and Lower sums) Let R be a rectangle in $\mathbb{R}^n$, $f : R \rightarrow \mathbb{R}$ be a bounded function and let P be a partition of R .

$$\mathcal{L}(P, f) := \sum_{S \in P(S \text{ is a subrectangle})} \left(\inf_{x \in S} f(x) \right) |S|$$

$$\mathcal{U}(P, f) := \sum_{S \in P(S \text{ is a subrectangle})} \left(\sup_{x \in S} f(x) \right) |S|$$

where $|S|$ is the volume of the subrectangle S . For convenience, we denote $\inf_{x \in S} f(x)$ by $m_S(f)$ and $\sup_{x \in S} f(x) = M_S(f)$.

Lemma A.2.3. For a bounded function $f : R \rightarrow \mathbb{R}$,

$$\mathcal{L}(P, f) \leq \mathcal{U}(P, f)$$

for a given partition P .

Proof. If S_i are subrectangles of S coming from P' , then $m_S(f)|S| = m_S(f)(|S_1| + |S_2| + \cdots + |S_i|) \leq \sum_{S_i \in S} m_{S_i}(f)|S_i|$. This proves the result for lower sums, and the proof for upper sums is similar.

□

Lemma A.2.4. If P' is a refinement of P (i.e. each subrectangle of P' is contained in some subrectangle of P) then

$$\mathcal{L}(P, f) \leq \mathcal{L}(P', f)$$

and

$$\mathcal{U}(P', f) \leq \mathcal{U}(P, f)$$

Appendix B

Exterior Forms

B.1 Definitions and Results

Definition B.1.1. (Exterior Forms) Let $k \geq 1$ and $n \in \mathbb{N}$. A k -form on $\mathbb{R}^n$ is a map

$$f : \underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{k \text{ copies}} \rightarrow \mathbb{R}$$

such that

1. f is linear in each of its k coordinates.
2. For every $\sigma \in \text{sym}(k)$, we have $f(x_{\sigma(1)}, x_{\sigma(2)} \cdots x_{\sigma(k)}) = \text{sgn}(\sigma)f(x_1, x_2, \cdots x_k)$.

Definition B.1.2. ($\Lambda^k(\mathbb{R}^n)$) We denote the space of all k -forms by $\Lambda^k(\mathbb{R}^n)$. By convention, we set

$$\Lambda^0(\mathbb{R}^n) = \mathbb{R}$$

Lemma B.1.3.

Proof.

□